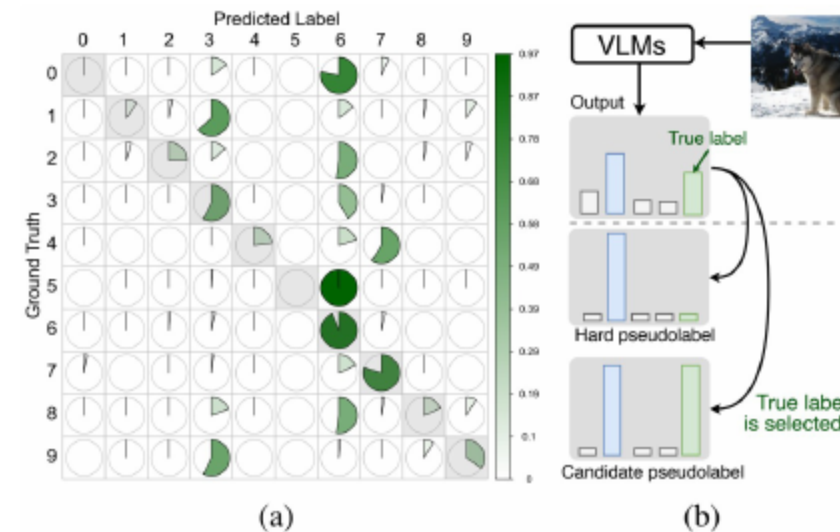


Candidate Pseudolabel Learning: Enhancing Vision-Language Models by Prompt Tuning with Unlabeled Data

Jiahan Zhang, Qi Wei, Feng Liu, Lei Feng

01 Why Hard Pseudolabels Fail For VLMs



- Zero-shot VLMs yield noisy, imbalanced hard pseudolabels
 - Dominant classes attract many wrong assignments
 - Minority classes are underrepresented, biasing training
- EuroSAT evidence shows severe misclassification skew
- Relying on top-1 labels destabilizes adaptation
 - Increases error reinforcement during fine-tuning
- Need unlabeled-data methods beyond

01 Problem Setting And Learning Paradigms

- Goal: adapt VLMs for C-class classification using unlabeled data
- Paradigms: semi-supervised, unsupervised, transductive zero-shot
- Leverage zero-shot predictions while mitigating noise and imbalance
- Compatibility with textual and visual prompt tuning required

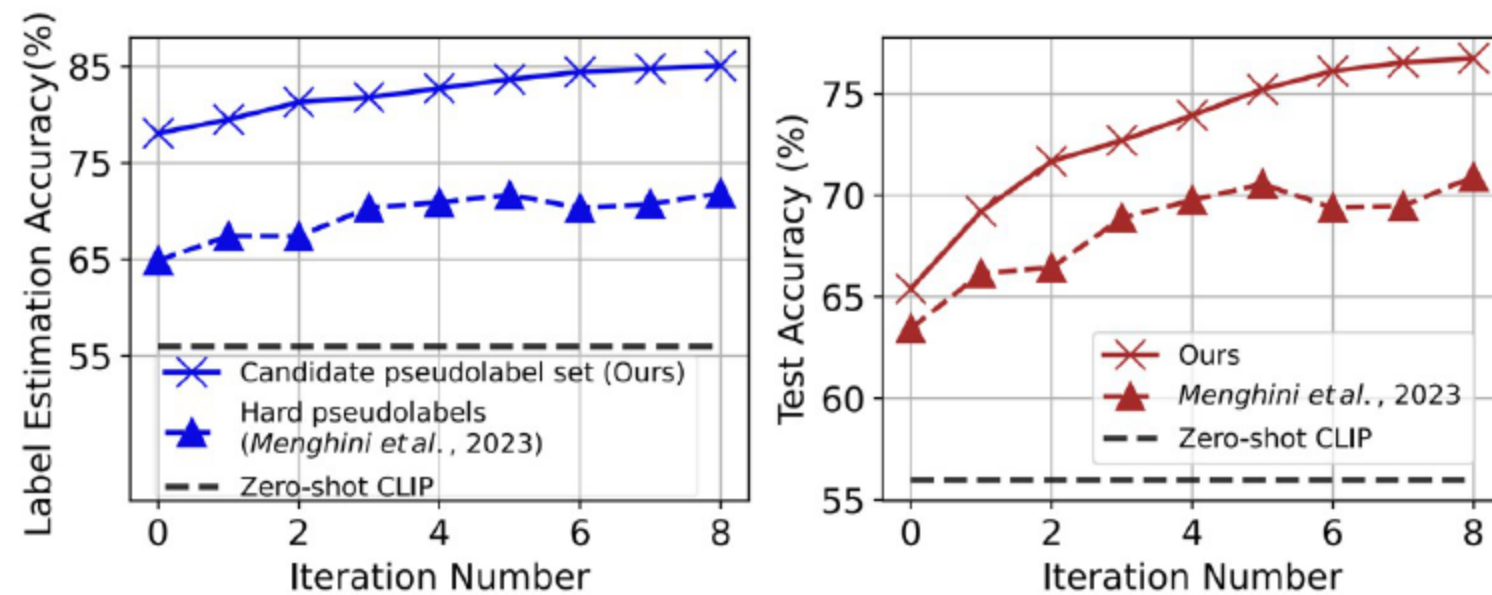
02 Related Work And Limitations

Vision-Language Models And Prompt Tuning

CLIP-like VLMs excel at zero-shot recognition
 Textual and visual prompt tuning offers parameter-efficient adaptation
 Existing approaches rely on reliable labels or robust pseudolabels
 Unlabeled settings often lack dependable supervision

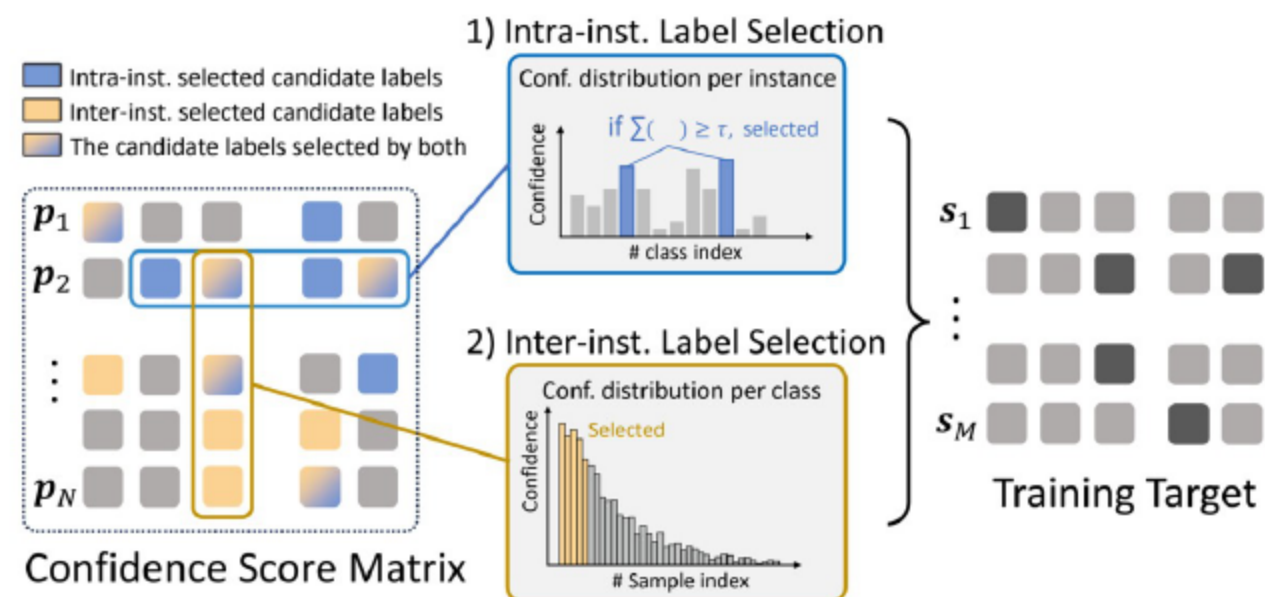
Learning From Unlabeled Data In VLMs

Classical pseudolabeling uses high-confidence selection
 VLM-specific methods refine hard labels or per-class confident sets
 When zero-shot accuracy is low, hard labels are brittle
 Methods can amplify class imbalance, motivating inclusive supervision



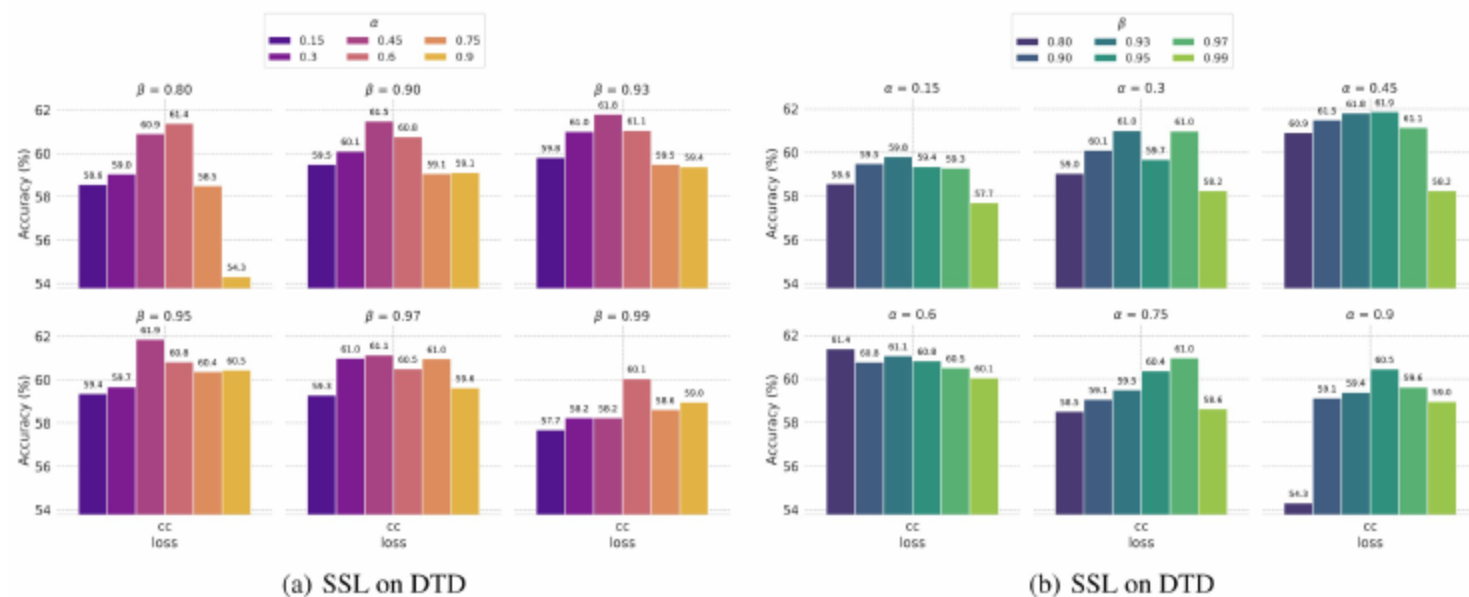
- Replace single hard labels with per-instance candidate label sets
- Use confidence matrix to drive intra- and inter-instance selection
- Convert to learning with candidate labels via partial-label losses

- Integrates with textual and visual prompt tuning
- Supports SSL, UL, and TRZSL paradigms
- Works with multiple partial-label losses and iterative refinement
- SOTA gains on nine datasets, strongest when zero-shot is weak

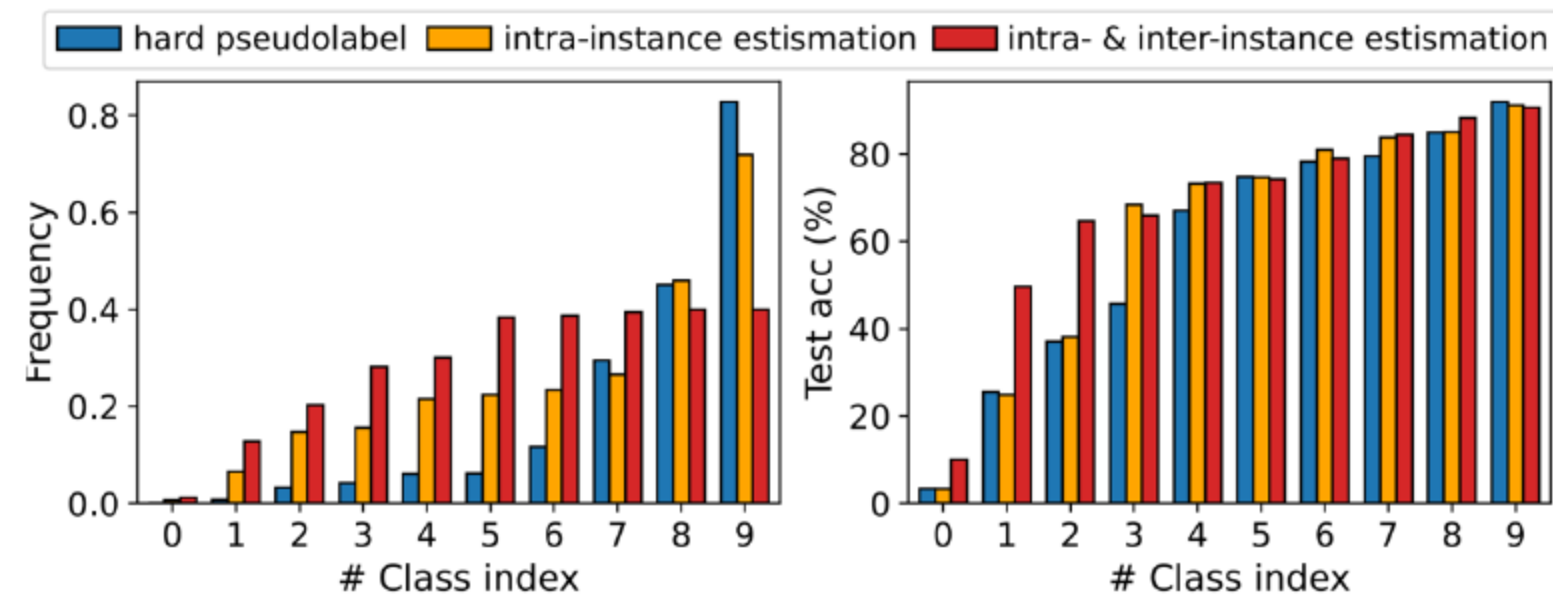


- Construct confidence matrix from per-instance class scores
- Intra-instance: adaptive top-K by cumulative confidence mass
- Inter-instance: per-class top- β fraction across data

- Transform labels to binary candidate vectors
- Optimize partial-label objectives; mix with CE if labeled data exist
- Iterate: update model, recompute confidences, refresh candidates
- Applicable to SSL, UL, and TRZSL settings



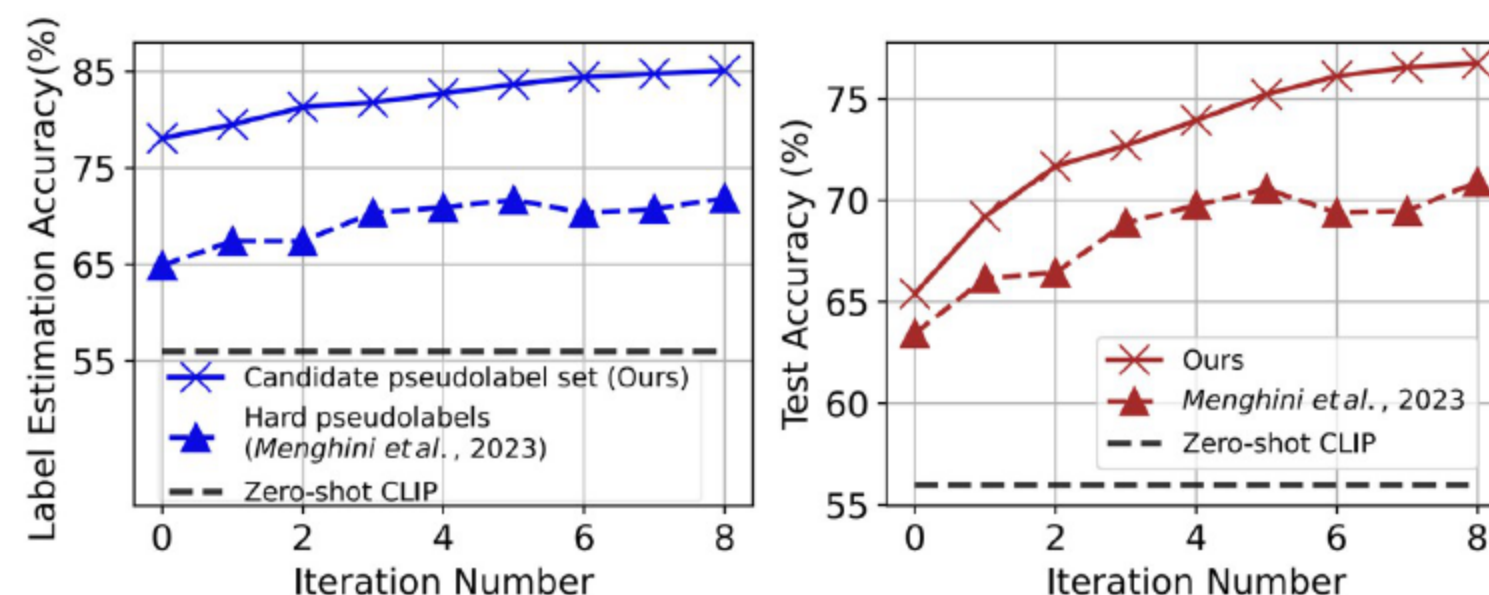
- Sort class confidences; include until cumulative mass exceeds τ
- τ adapts via α -quantile of per-instance maxima per iteration
- Improves true-label coverage without large candidate sets



- For each class, select top- β confidences across instances
- Counters dominance of easy classes; balances candidate frequency
- Final candidate set is intersection with intra-instance selection

Methods	EuroSAT	DTD	Flowers102
CPL _{CC}	67.26	51.91	72.90
CPL _{RC}	66.38	51.58	71.91
CPL _{CAV}	66.91	49.31	72.06
CPL _{LW}	67.15	52.23	72.33
CPL _{Soft CE}	56.85	49.53	71.87
CPL _{Soft CC}	65.58	50.98	72.58
CPL _{Soft RC}	65.89	50.83	71.89
CPL _{Soft CAV}	64.90	48.78	71.72
CPL _{Soft LW}	65.74	51.04	72.15

- Train on instances with non-empty candidate sets
- Use candidate-aware losses (CC, RC, CAV, LW, soft CE)
- Mix labeled CE and candidate loss in SSL/TRZSL



- Periodically recompute confidence matrix and update τ
- Regenerate candidates via both selections; continue training
- Improves label inclusion, reduces bias, raises accuracy

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Benchmarks, Paradigms, And Tuning Setups

	Flowers102	RESISC45	DTD	CUB	EuroSAT	FGVCAircraft
Statistic data						
Class number	102	45	47	200	10	100
Training set size	2040	6300	3760	5594	27000	6667
Testing set size	6149	25200	1880	5794	5000	3333
Training procedure						
Network	ViT-B / 32					
Batch size	64					
Epoch	50 where first two epochs are set for warmup					
Optimizer	SGD					
Momentum	0.9					
Learning rate (LR)	0.02					
Weight decay	5e-2					
LR scheduler	CosineAnnealingLR					
Hyperparameters						
α in intra-instance label selection	0.60	0.90	0.75	0.75	0.75	0.90
β in inter-instance label selection	0.99	0.97	0.95	0.99	0.80	0.97

- Nine datasets across scenes, textures, fine-grained, and actions
- Evaluate SSL, UL, TRZSL with textual and visual prompts
- Default ViT-B/32; iterative training with SGD cosine decay

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Overall Performance Gains

	Flowers-102	UCF-101	CIFAR-100	EuroSAT	DTD	CALTECH-101
CLIP	66.6	61.0	64.2	45.1	42.9	90.5
CLIP-PR	57.7	57.9	63.2	44.2	40.1	84.8
UPL	71.5	63.9	65.8	62.2	48.0	90.6
LaFTer	71.0	68.2	74.6	73.9	46.1	93.3
LaFTer + Ours	76.7	71.0	77.3	82.2	56.3	93.4

- CPL outperforms hard-label baselines across datasets and paradigms
- Largest gains where zero-shot CLIP is weak
- Better utilization of unlabeled data

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Baselines And Metrics

Methods	Flowers102			RESISC45			DTD		
	SSL	UL	TRZSL	SSL	UL	TRZSL	SSL	UL	TRZSL
Zero-shot CLIP	63.67 _{0.00}			54.48 _{0.00}			43.24 _{0.00}		
FPL	75.96 _{0.74}	65.67 _{0.23}	80.97 _{0.00}	68.13 _{0.55}	63.07 _{0.38}	72.11 _{0.00}	37.10 _{5.45}	44.96 _{0.55}	46.30 _{0.03}
CPL (Ours)	77.36 _{0.24}	70.01 _{0.21}	84.60 _{0.10}	71.73 _{0.57}	68.47 _{0.34}	72.16 _{0.26}	54.63 _{0.79}	48.92 _{0.17}	59.79 _{1.32}
GRIP	83.60 _{0.48}	69.84 _{1.06}	86.26 _{0.00}	74.11 _{0.68}	70.55 _{0.88}	81.07 _{0.00}	56.07 _{0.85}	46.09 _{1.06}	65.30 _{0.01}
CPL (Ours)	89.66 _{0.36}	72.90 _{0.78}	87.35 _{0.76}	80.98 _{0.11}	77.39 _{0.44}	85.85 _{0.49}	61.21 _{0.56}	51.91 _{0.71}	68.00 _{0.34}
Methods	CUB			EuroSAT			FGVCAircraft		
	SSL	UL	TRZSL	SSL	UL	TRZSL	SSL	UL	TRZSL
Zero-shot CLIP	51.82 _{0.00}			32.88 _{0.00}			17.58 _{0.00}		
FPL	55.29 _{0.59}	53.04 _{0.53}	55.44 _{0.20}	62.05 _{1.64}	48.96 _{1.49}	53.70 _{26.87}	20.02 _{0.77}	16.62 _{0.67}	17.55 _{0.37}
CPL (Ours)	56.37 _{0.45}	54.18 _{0.05}	64.01 _{0.17}	64.84 _{2.15}	51.45 _{1.97}	54.03 _{2.27}	22.37 _{0.66}	18.90 _{0.20}	28.47 _{0.43}
GRIP	56.65 _{0.33}	51.42 _{0.21}	59.48 _{0.38}	58.66 _{2.64}	57.21 _{1.77}	92.33 _{0.69}	16.98 _{0.82}	15.22 _{0.71}	26.08 _{0.25}
CPL (Ours)	58.53 _{0.24}	53.47 _{0.36}	66.20 _{0.50}	77.51 _{0.80}	67.26 _{0.47}	93.78 _{0.12}	22.48 _{0.63}	18.35 _{0.27}	30.86 _{0.70}

- Compare against FPL, GRIP, CLIP-PR, UPL, LaFTer
- Report top-1 accuracy; TRZSL uses harmonic mean
- Analyze robustness, losses, imbalance, encoder size, data-efficiency

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Versatility And Loss Function Robustness

	Flowers-102	UCF-101	CIFAR-100	EuroSAT	DTD	CALTECH-101
CLIP	66.6	61.0	64.2	45.1	42.9	90.5
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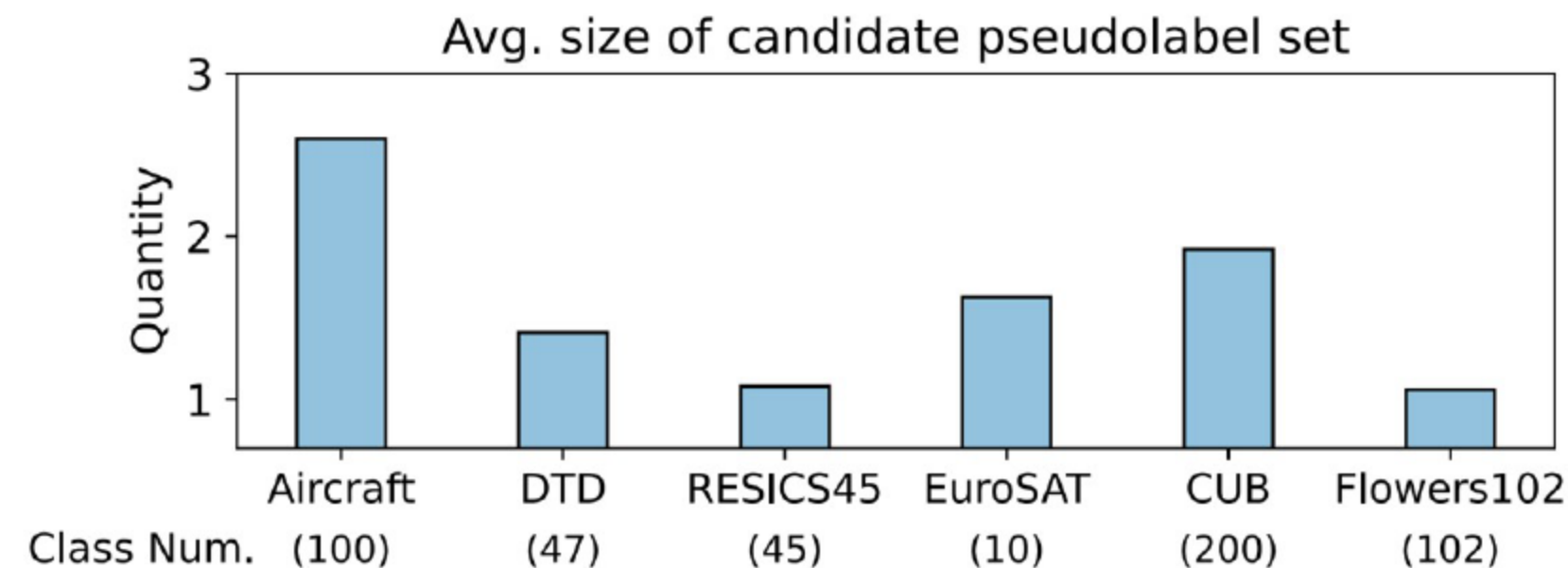
- CPL plugged into LaFTer yields best results on many benchmarks
- Strong performance across CC, RC, CAV, LW, soft CE
- Equal treatment of candidates mitigates bias better than soft targets

07 Class Imbalance, Encoder Size, And Data Usage

Methods	Balanced	Imbalanced $\delta=100$	Imbalanced $\delta=50$
LaFTer	74.64	65.63	66.59
CPL (w/o inter)	76.07	66.68	67.85
CPL	77.32	67.70	69.65

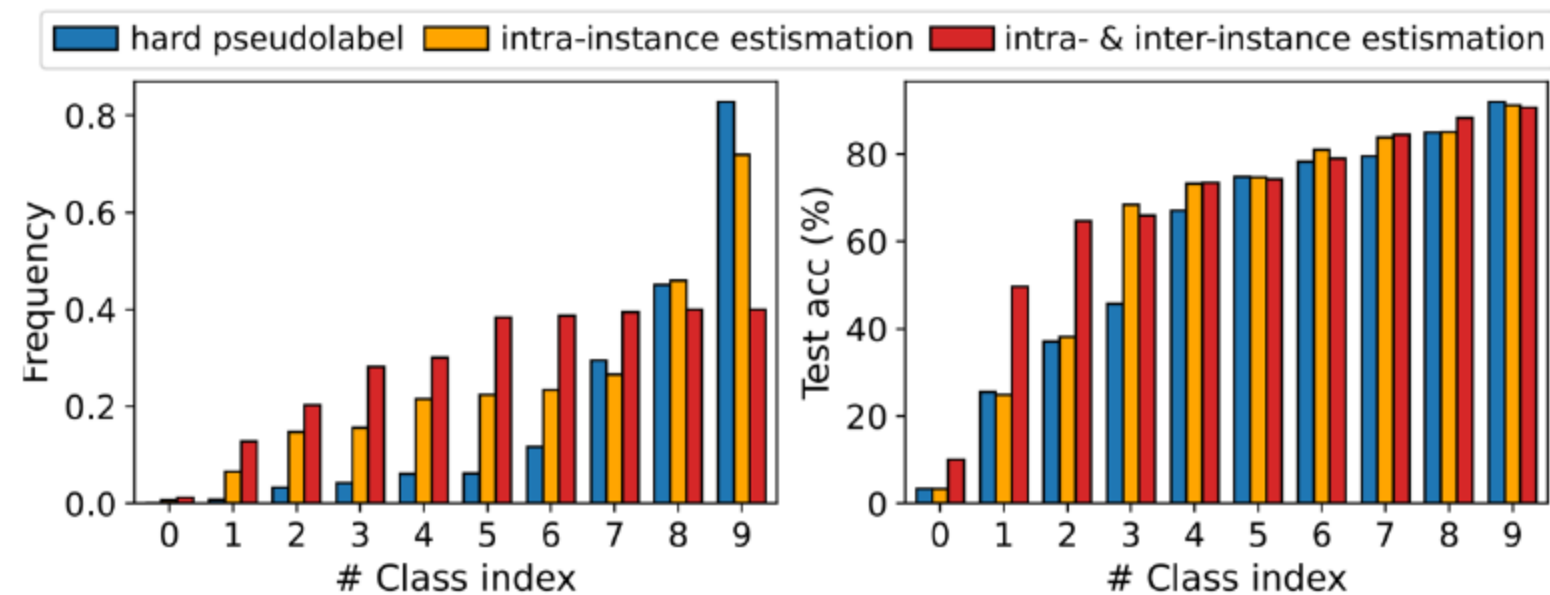
- Inter-instance selection is crucial on imbalanced data
- CPL scales with larger encoders (ViT-L/14)
- Performance saturates with more unlabeled data

07 Candidate Set Size And Hyperparameters



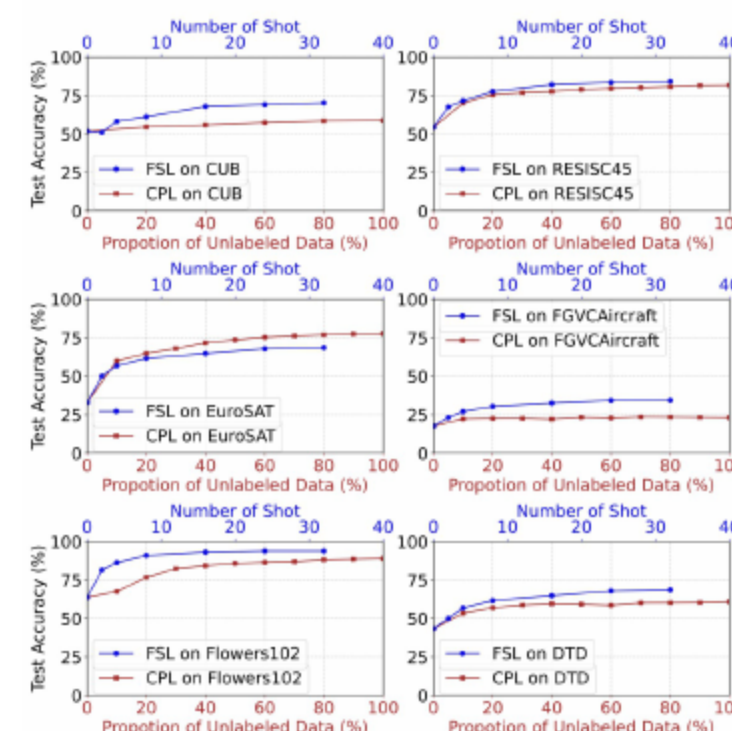
- Average candidate set sizes near 1, limiting ambiguity
- Best $\alpha \approx 0.45-0.75$; $\beta \approx 0.95-0.97$; very high β reduces instances
- Both selections improve class balance and per-class accuracy

08 Effect Of Each Selection Strategy



- Intra-instance selection improves balance and accuracy over hard labels
- Adding inter-instance selection further equalizes distribution
- Intersection yields high-quality, balanced candidates

08 When To Prefer Labeled Versus Unlabeled



- Improvements saturate sooner on fine-grained, many-class datasets
- Few high-quality labels can outperform massive unlabeled data
- Allocate resources accordingly under constraints

- Depends on candidate quality; true labels may be excluded
- Future: smarter thresholds, better candidates, uncertainty modeling
- Target robustness under domain shift and imbalance

- CPL offers balanced, high-coverage candidate pseudolabels for VLM adaptation
- Combines adaptive selections, partial-label losses, iterative refinement
- Delivers consistent SOTA, especially where zero-shot is weak



THANKS!